

Human-Computer Interaction A1 (2025/2026)

Team Name

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1 Introduction

1.1 Team Members

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1.2 Domain of Interest

Theme 3: Sustainable tourism

Our team chose the domain of environmentally sustainable tourism, focusing on promoting environmentally friendly travel habits such as walking, using public transport, and focusing on local activities instead of international ones, to reduce pollution and support communities. We found it interesting because tourism often harms the environment, and we wanted to explore how HCI can encourage greener behaviors.

To increase sustainability, travelers can adopt responsible habits and use tools that guide sustainable choices. Like simulating graphic of travel methods and calculating stats, how is different to move from one point to another, buying tickets online reduce use of paper and print, eating from local instead of supply chain of fast foods (limiting international transportation), waste reduction: carrying reusable bottles, avoiding single-use plastics, and proper recycling...

People are somewhat aware of environmental issues, but many do not act consistently. Digital and interactive designs can help raise awareness by showing the environmental impact and making more environmentally friendly decisions.

To make them more aware and motivated, we can use gamification — for example:

- Earning points or badges for eco-friendly actions (like using green transport).
- Completing challenges or missions such as “Eat Local for a Day.”
- Leaderboards: Tracking progress and sharing achievements with others.
- Walking for 3 km for a day
- Renting an electric car if want high mobilities between places
- Consuming foods in place instead of take away

This makes sustainable tourism more interactive, fun, and rewarding, helping people travel responsibly while enjoying the experience.

2 Methodology

2.1 Participants

- **Participant 1: Immediate user**

- Name: Carafassi Samuele
- Occupation: Cybersecurity student at the University of Padua

Why was he chosen?

- University student.
- Comfortable using technology and travel apps.

Why is he appropriate?

- Represents the lifestyle of active young users.

Recruitment method:

- Contacted through the University of Padua.
- Informed about the study's purpose and consented.

- **Participant 2: immediate user**

- Name: Farinelli Simone
- Occupation: Game develop and design student at the University of Bristol

Why was he chosen?

- University student.
- Lives abroad, always on travelling.
- Comfortable using technology.

Why is he appropriate?

- Has experience with digital tools for travel and mobility.
- Can provide feedback relevant to student and travel contexts.

Recruitment method:

- Contacted through the University of Bristol.
- Informed about the study's purpose and consented.

- **Participant 3: Extreme User**

- Name: Isa Yaqoubi
- Occupation: Biomedical engineering students at the University of Bologna

Why was he chosen?

- Unibo student.

- Comfortable using technology and travel apps.

Why is he appropriate?

- Has experience with digital tools for travel and mobility.
- Dynamic mindset.

Recruitment method:

- Recommend from a friend.

• **Participant 4: Immediate user**

- Name: Ramzy cherif
- Occupation: Recently graduated and working at the National Erasmus+ Office.

Why was he chosen?

- Constant communication with Erasmus students.

Why is he appropriate?

- Can provide feedback relevant to travel contexts.

Recruitment method:

- Contacted through a friend.

2.2 Interview Context

To better understand travelers' habits, motivations, and awareness about environmentally sustainable tourism, we plan to conduct interviews and observational studies with participants who travel for leisure or work. The goal is to explore how people make travel decisions, what factors influence their choices, and how technology or gamified systems could encourage more sustainable behavior. The tasks are done through physical and remote interviews.

2.3 Interview and Observation Details

The questions will start broadly (about general travel habits and preferences) and gradually move toward specific actions related to transport, food choices, and environmental awareness.

Open Questions

1. Can you describe your last trip and what made it memorable?
Why: This opens the conversation and allows you to observe their natural travel habits without leading them.
2. When you plan a trip, what are the most important factors for you?
Why: Establishes their priorities. You can later see if "sustainability" is mentioned organically.
3. How do you typically get around when you're in a new city? What makes you choose that option?
Follow-up: Have you ever considered walking, biking, or using public transport instead? What encouraged or discouraged you?
4. Tell me about your eating habits when you travel. How do you choose where to eat?
Follow-up: Do you consider eating at local, non-chain restaurants? What are the pros and cons for you?
5. What tools or apps do you use when you travel? Have you ever used one to help you make sustainable choices?
Follow-up: If yes, what did you like or dislike about it? If no, why not? How do you usually handle tickets, bookings, or payments when you travel? (digitally or on paper?)
Why: Understands the competitive landscape and existing user mental models.
6. What kinds of things do you usually bring with you to make traveling easier (like bottles, bags, gadgets)?
7. What does 'sustainable tourism' mean to you? Does it influence your travel decisions?
Why: Gauges their baseline awareness and personal definition of the concept.
8. Have you ever thought about how your choices — like transport, food, or accommodation — might affect the places you visit?
9. How would you feel about using an app that tracks your travel choices and gives feedback or rewards for certain actions?
10. If a system gave challenges or badges for things like walking, trying local food, or reducing waste, would you find it motivating? Why or why not?
11. If you could design an ideal tool or service that helps you travel easier, meanwhile maintaining sustainably, what would it do?
Why: Lets users imagine solutions — great for concept validation.

Scale-based Questions

1. Traveling is an important and regular part of my lifestyle.
0 (Not important) → 5 (Very important)
2. I enjoy figuring out and using local public transportation when I travel
0 (Not at all) → 5 (Very much)
3. I consistently choose to buy e-tickets or make online bookings to avoid using paper.
0 (Never) → 5 (Always)
4. I actively seek out local restaurants, markets, and tours over international chains.
0 (Never) → 5 (Always)
5. I prefer exploring quiet, local neighborhoods rather than crowded tourist hotspots.
0 (Disagree) → 5 (Agree)

6. I would be interested in a tool that shows me the environmental impact of my different travel options.
0 (Not interested) → 5 (Very interested)
7. I would find it motivating to compete with friends on a leaderboard for completing eco-friendly travel actions.
0 (Not motivating) → 5 (Very motivating)
8. I consciously plan my travel routes to minimize my environmental footprint.
0 (Never) → 5 (Always)

Conclusive Question

Do you have any thoughts or suggestions about sustainable tourism?

2.4 Roles

- **Observer:** OBS
- **Note-taker:** Jiahao Guo
- **Interviewer:** Kalile Horri

2.5 Materials Used

- **PC:** Main device used to host and manage the recording and note-taking.
- **Google Meet:** Used for conducting and recording the online meeting/interview.
- **OBS:** Utilized to record and manage screen/audio captures during the session.
- **Trello:** Used as a digital notebook to organize and take notes throughout the process.

3 Results

3.1 Key Quotes

Participant 1 Summary:

The participant demonstrates moderate awareness of sustainable tourism but does not consistently act on it. Travel decisions are primarily influenced by *cost, convenience, and planning*, rather than environmental impact.

He typically walks short distances, uses public transport, or rents a car depending on distance and flexibility needs. Public transport is recognized as greener but may be limited by delays or restrictions.

Food choices are mostly *price-driven*, with occasional interest in local restaurants if they appear interesting or affordable.

The subject uses digital tools such as Google Maps and search engines for planning, but has not actively used apps or tools to support sustainable decisions. Minimal sustainable practices are currently adopted (e.g., paperless bookings, reusable bottles), and sustainability is not a personal priority.

Motivation for eco-friendly actions increases when *gamification, achievements, or rewards* are involved. Weak or indirect rewards are not effective, while badges, challenges, or leaderboards are highly engaging.

In Samuele's opinion, data tracking for travel suggestions is considered negatively, as he is concerned about the collection of personal information. Overall, the participant shows openness to sustainable travel when it is convenient, interactive, and provides tangible or social incentives.

Scale Responses:

- Travel importance: 5/5
- Enjoy using public transport: 4/5
- Consistently use e-tickets: 5/5
- Seek local restaurants/markets: 3/5
- Prefer local neighborhoods over tourist hotspots: 3/5
- Interested in environmental impact tools: 2/5
- Motivated by eco-friendly leaderboards: 3/5
- Plan routes to minimize environmental footprint: 4/5

Participant 2 Summary

The participant visited Valencia in 2023 and expressed appreciation for the city's food, beauty, and futuristic infrastructure, especially its transport system. Travel decisions are primarily guided by affordability, convenience, and overall *simplicity and sustainability or environmental impact*.

He favors affordable options for food and accommodation, emphasizing good value and safety. Cultural attractions such as museums and natural spaces are important aspects of his travel experience. *Simplicity and user reviews* play a central role in his decision-making process.

For transportation, he prefers using the metro as an affordable and convenient option, as well as walking for shorter distances. Buses are used occasionally but not favored (because of frequent delays), while bicycles are actively avoided (for personal issue).

He makes use of local transport systems and relies heavily on *digital applications for navigation and planning*.

Digital tools such as Google Maps, Skyscanner, regional trainline apps, TripAdvisor, and local websites are essential in his travel planning process. He prefers an entirely *digital approach*, resorting to paper only when strictly necessary. Phone-based solutions are the main tools he uses.

The participant shows limited but emerging awareness of sustainable tourism. He values the idea of integrating tourism with local culture and maintaining the ecosystem, even if he has not deeply reflected on it. His travel behavior demonstrates respect for local living, avoidance of taxis and car rentals, and openness to sustainable alternatives when convenient.

Motivation for eco-friendly choices increases when supported *by social feedback or tangible benefits*. The participant appreciates feedback systems and interactive challenges that enhance engagement. However, he prefers self-improvement approaches rather than competitive ones.

In Simone's opinion, the importance of preserving natural environments and ensuring that travel paths remain safe and respectful of local ecosystems is fundamental. He values nature's beauty and believes that tourism should actively prevent negative impacts on local communities and fauna.

Although sustainability is not yet fully embedded in his personal behavior, he identifies these aspects as essential aspects of green tourism.

Scale Responses:

- Travel importance: 4/5
- Enjoy using public transport: 5/5
- Consistently use e-tickets: 5/5
- Seek local restaurants/markets: 2/5, It depends by country. (British foods are inedible, Spanish cuisine is awesome, etc...)
- Prefer local neighborhoods over tourist hotspots: 3/5
- Interested in environmental impact tools: 4/5
- Motivated by eco-friendly leaderboards: 2/5
- Plan routes to minimize environmental footprint: 3/5

Participant 3 Summary

The participant recently visited Southern Italy with his brother after a long exam period, describing the trip as both relaxing and memorable.

When planning a trip, the participant prioritizes *budget-friendly options*, emphasizing that financial limitations as a student strongly influence his decisions. Accommodation is chosen based on affordability, as he prefers to spend more time exploring rather than focusing on comfort. He typically opts for *cheap lodging and public transportation*, maintaining that the essence of travel lies in the experience of discovering new places rather than material aspects.

In terms of mobility, he prefers *walking whenever possible*, often selecting accommodation in the city center to reduce reliance on transport. During his last trip, he explored entirely on foot, valuing the opportunity to immerse himself in the local atmosphere. Public transportation is used occasionally but primarily for *environmental and cost-saving reasons*.

Regarding food choices, the participant consistently seeks *local dishes and traditional cuisine*, considering gastronomy a key part of cultural discovery. He often asks locals for recommendations to experience authentic flavors. International chains are consciously avoided, as he perceives them as incompatible with the spirit of travel.

He relies heavily on *digital tools* such as ChatGPT and Google Maps to organize travel, find affordable restaurants, and receive real-time suggestions. Digital payments and *e-tickets* are strongly preferred over paper options, which he views as both more convenient and environmentally friendly.

The participant also demonstrates *sustainable travel habits*, carrying reusable items such as a water bottle, a personal glass, and a hiking backpack to minimize plastic use. He defines sustainable tourism as traveling with minimal impact on the environment and protecting nature for future generations.

He expressed interest in a *mobile application* that promotes sustainable travel by tracking actions, giving feedback, and suggesting eco-friendly options. He found the concept of *gamified challenges and achievement badges* particularly appealing, considering it an enjoyable way to motivate travelers. However, he prefers *self-motivation and intrinsic satisfaction* over competition with others.

Scale Responses:

- Travel importance: 5/5
- Enjoy using public transport: 4/5
- Consistently use e-tickets: 4/5
- Seek local restaurants/markets: 5/5
- Prefer local neighborhoods over tourist hotspots: 4/5
- Interested in environmental impact tools: 4/5
- Motivated by eco-friendly leaderboards: 3/5
- Plan routes to minimize environmental footprint: 5/5

Participant 4 Summary

The participant recently traveled to Italy to visit an old friend after several years apart, describing the trip as a memorable experience for the opportunity to reconnect and discover new traditions and lifestyles. He emphasized that meeting his friend and exploring a different culture made the journey truly special.

When planning a trip, the participant focuses on *organization and preparation*. He carefully books tickets, packs clothes and food, reserves hotels, and plans daily activities in advance. Dietary restrictions also influence his choices, as he needs to ensure that suitable meals are available at his destination. He values *structured planning* and enjoys creating detailed schedules for his travels.

In terms of mobility, he reads *articles about the city and its weather conditions* before traveling, as he prefers mild climates and calm environments. He enjoys *biking* as a way to explore new areas and considers it both healthy and enjoyable. Public transportation is also an option, though he often chooses to rent bikes when possible.

Regarding food choices, the participant prefers *healthy meals* but remains curious to try local dishes. Food is a central element of his cultural exploration, and he enjoys learning about the origins and preparation of traditional meals.

He uses *digital tools* such as Booking.com to manage reservations, guided by reviews and star ratings. He expresses high satisfaction with digital booking systems and prefers *online payments and e-tickets* for their convenience and reliability.

The participant travels with well-organized luggage, separating clothing, personal care, and food items. He carries a *metal water bottle* to reduce plastic use and always brings his *laptop, phone, and tablet* to read, work, and stay productive while traveling. These habits reflect both practicality and eco-consciousness.

He defines *sustainable tourism* as environmentally friendly and economically viable travel that respects cultural and social contexts. He believes that respecting nature and communities benefits everyone, expressing that “if you don’t respect the environment, it won’t respect you either.”

The participant reacted positively to the idea of a *mobile application* that tracks travel behavior and provides rewards for eco-friendly actions, noting that privacy and data security must be prioritized. He believes such tools could save time, offer useful suggestions,

and create a mutually beneficial experience.

He finds *gamified challenges and reward systems* highly motivating, viewing competition as a fun and effective way to encourage sustainable travel. He believes that even without winning, eco-friendly behavior is its own reward.

Overall, the participant considers sustainable tourism a “*win for everyone.*” He encourages travelers to adopt eco-conscious habits and believes that small, mindful actions—such as biking, choosing local food, and reducing waste—make trips more meaningful and environmentally responsible.

Scale Responses:

- Travel importance: 4/5
- Enjoy using public transport: 3/5
- Consistently use e-tickets: 4/5
- Seek local restaurants/markets: 2/5
- Prefer local neighborhoods over tourist hotspots: 5/5
- Interested in environmental impact tools: 4/5
- Motivated by eco-friendly leaderboards: 5/5
- Plan routes to minimize environmental footprint: 4/5

4 User Needs

4.1 User Needs

1. Needs access to travel bundles that offer convenient, integrated transportation options
2. Motivational rewards for engaging in eco-friendly actions
3. Transparent data collection practices
4. Price-driven: decisions depend on the value and offerings available, if it is sustainable then will be great
5. Access to reliable eco-friendly transport information
6. Safe, simple and low-impact travel routes
7. Positive, non-competitive sustainability feedback
8. Connection to local eco-initiatives
9. Affordable travel solutions that prioritize exploration over comfort
10. Easy access to local and authentic food and cultural recommendations
11. Digital, paperless tools for planning and navigating trips efficiently
12. Self-motivating sustainability features with personal progress tracking rather than competition
13. Tools that support well-organized travel planning, including schedules, reservations, and dietary considerations
14. Reliable access to healthy and local food options that fit personal restrictions
15. Gamified and competitive sustainability rewards to motivate eco-friendly choices
16. Trusted digital systems with clear privacy and data security for bookings and tracking

4.2 Deep User Needs

- **Make sustainable choices easy:** Users want eco-friendly options to be simple and convenient during travel, without causing extra effort or obstacles.
- **Feel safe and in control:** They need clear information to plan confidently, stay within budget, and handle personal needs.
- **Trust and transparency:** Users want to know that their data is protected and that the app is honest about sustainability.
- **Connect with real local experiences:** They want to explore culture and food in an authentic way, not tourist traps.
- **Stay motivated in a friendly way:** They like rewards and feedback, but don't want stressful competition.

4.3 Brainstorm artifact

4.3.1 Make sustainable choices easy

- Automatically highlight eco-friendly travel options first
- Provide short “green travel shortcuts” for common routes
- Offer simple travel bundles combining low-impact mobility choices
- Use clear icons/labels to show environmental impact instantly
- Suggest actions that require zero extra effort (ex: walking paths already near them)

4.3.2 Feel safe and in control

- Show travel confidence indicators (safe routes, well-lit areas, weather suitability)
- Include support for personal constraints (diet, accessibility, physical limits)
- Provide clear preparation checklists before and during the trip
- Offer calm, step-by-step guidance instead of overwhelming maps
- Allow users to preview experiences (crowd level, distance, difficulty, costs)

4.3.3 Trust and transparency

- Explain data usage in simple, transparent language
- Show how user actions directly benefit local nature and culture
- Let users choose what data to share and when
- Allow anonymous usage where possible (less personal data needed)

4.3.4 Connect with real local experiences

- Highlight small local businesses and authentic cultural spots
- Connect travelers with community-hosted experiences (markets, events)
- Suggest food based on local traditions, not global chains
- User recommendation system: they can share cuisine experiences between posts.

4.3.5 Stay motivated in a friendly way

- Provide personal progress journeys instead of rankings
- Use friendly encouragement and reflection prompts
- Offer optional playful challenges
- Highlight the positive impact of each small action taken
- Celebrate everyday actions (refill bottle, choose walking)

5 Solutions

5.1 Chosen Solution

The proposed solution is a **Smart Sustainable Travel Companion** that guides users to make eco-friendly choices while exploring safely and enjoying authentic local experiences. It addresses the needs to *make sustainable choices easy, feel safe and in control, trust the system*, and *connect with local culture* by providing clear environmental indicators, safety information, transparent data usage, and motivational feedback. Users can share their experiences through a **recommendation system**, receive **friendly encouragement and reflection prompts**, complete **optional playful challenges**, and see the **positive impact of each small action**, celebrating everyday sustainable choices such as refilling bottles or walking instead of using transport. This solution was selected because it integrates planning, exploration, cultural engagement, and reflection into a seamless experience, supporting sustainable behaviors without extra effort or stress.

5.2 Selection Rationale

This solution was chosen through a team voting process and by intersecting identified user needs with analyzed deep needs. It was selected because it effectively integrates planning, exploration, cultural engagement, and reflection into a seamless experience, supporting sustainable behaviors without extra effort or stress.

6 Project Name and Value Proposition

6.1 Project Name

Our team decided on the name **EcoGo** to reflect the core values of our solution. “Eco” emphasizes the focus on sustainability and environmentally friendly travel, while “Go” highlights mobility, exploration, and the ease of using the system during trips.

6.2 Value Proposition

if you don't respect the environment, it won't respect you either